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(54) **DC VOLTAGE FILTER FOR AN INPUT OF AN INVERTER OF AN ELECTRIC MACHINE FOR PROPELLING AN AIRCRAFT**

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(58) **Field of Classification Search**
CPC H02M 1/14; B60L 220/10; B60L 2210/42; B64D 27/30; B64D 27/34; B64D 27/35; B64D 27/359
See application file for complete search history.

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(57) **ABSTRACT**

A filter includes a positive busbar and a negative busbar that have a positive terminal and a negative terminal, respectively; at least one positive output terminal and at least one negative output terminal, respectively; and two portions, respectively, each portion having parallel folds to define planes which succeed one another. Each plane is inclined relative to the next plane. One of the portions is an internal portion that is fitted inside the other portion, which is an external portion. Each plane of one portion extends in parallel with and facing an associated plane of the other

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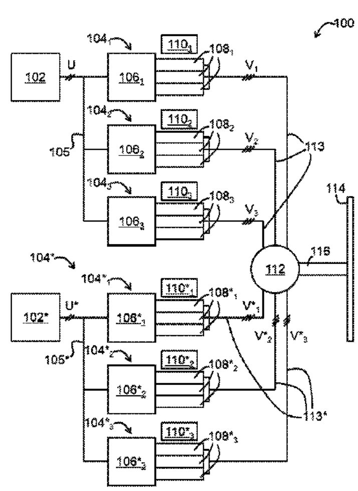

portion. Differential mode capacitors connected between the two portions.

8 Claims, 9 Drawing Sheets

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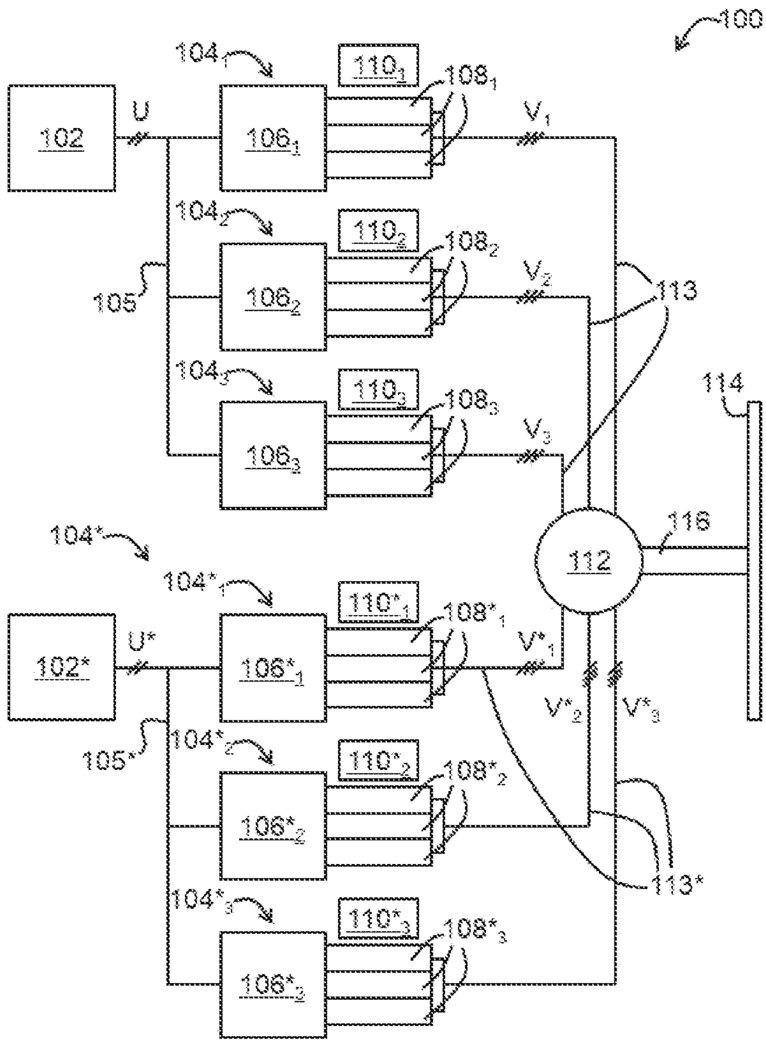
100
102
U
104_1
106_1
110_1
108_1
V_1
104_2
106_2
110_2
108_2
V_2
105
104_3
106_3
110_3
108_3
V_3
113
114
116
112
104*
104^*_1
102*
U*
106^*_1
110^*_1
108^*_1
V^*_1
V^*_2
V^*_3
105*
104^*_2
106^*_2
110^*_2
108^*_2
113*
104^*_3
106^*_3
110^*_3
108^*_3

FIG. 1

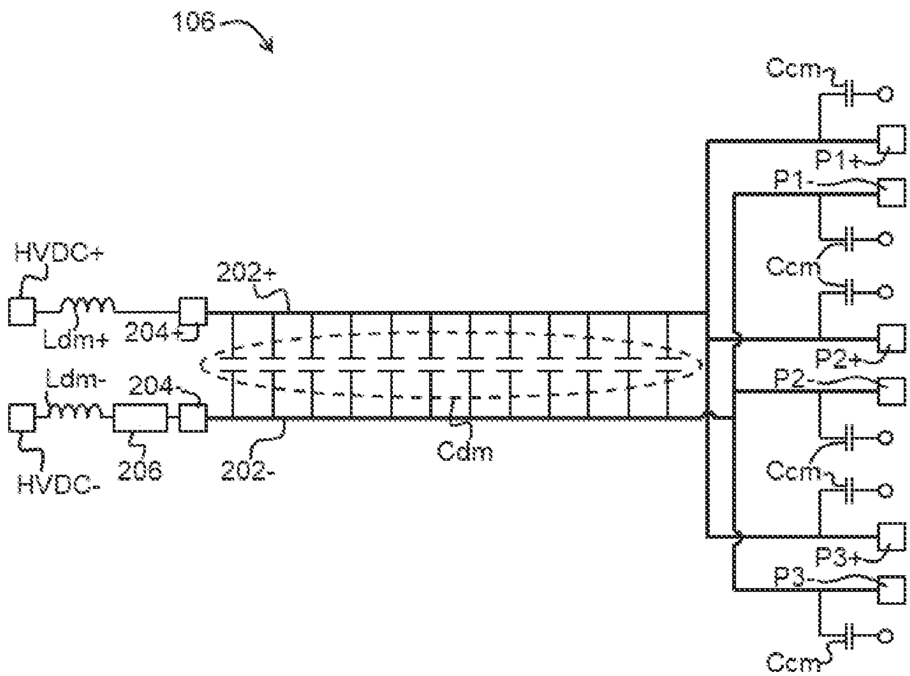
106
Ccm
P1+
P1-
HVDC+
202+
Ccm
Ldm+
204+
P2+
Ldm-
204-
P2-
HVDC-
206
202-
Cdm
Ccm
P3+
P3-
Ccm

FIG. 2

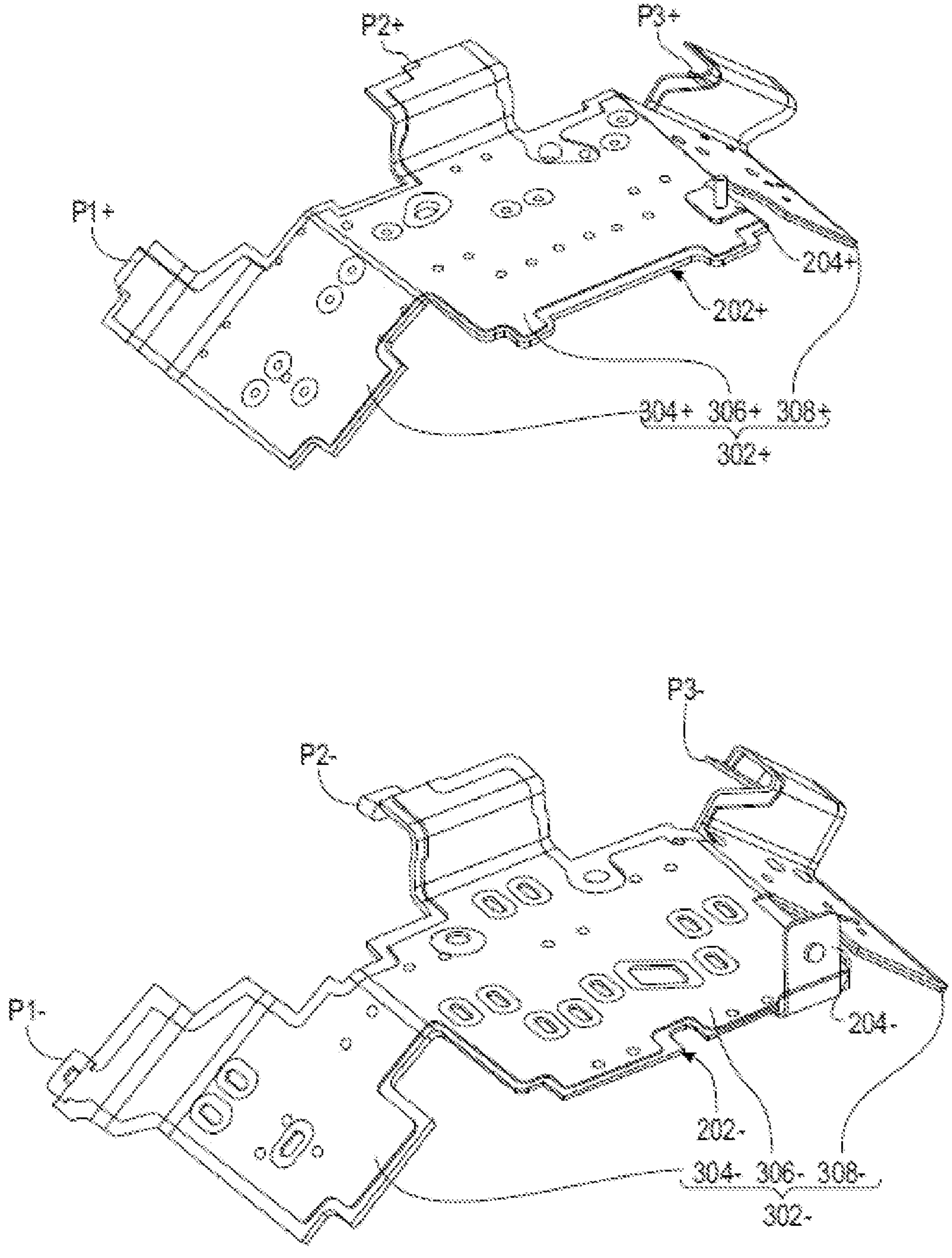

FIG. 3

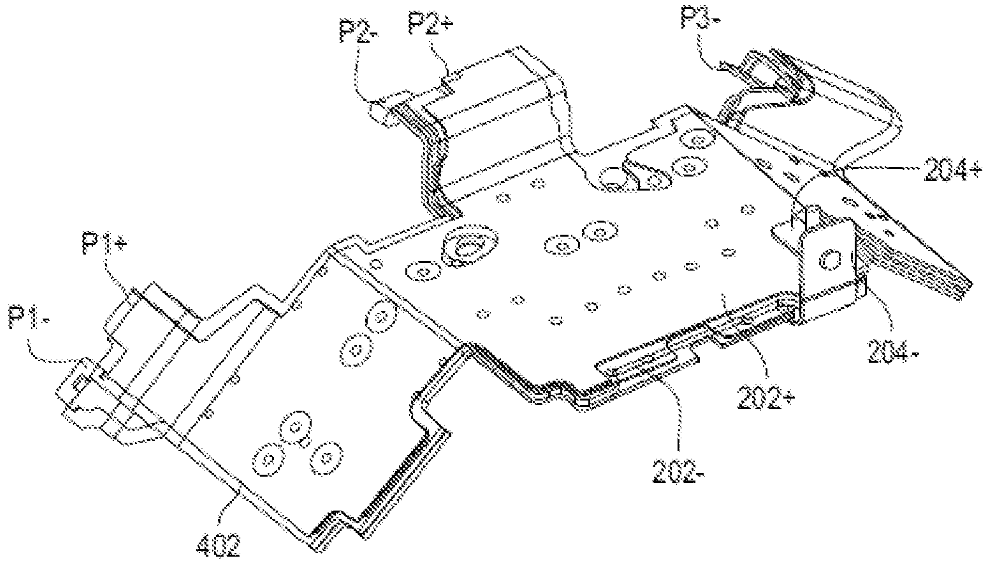

FIG. 4

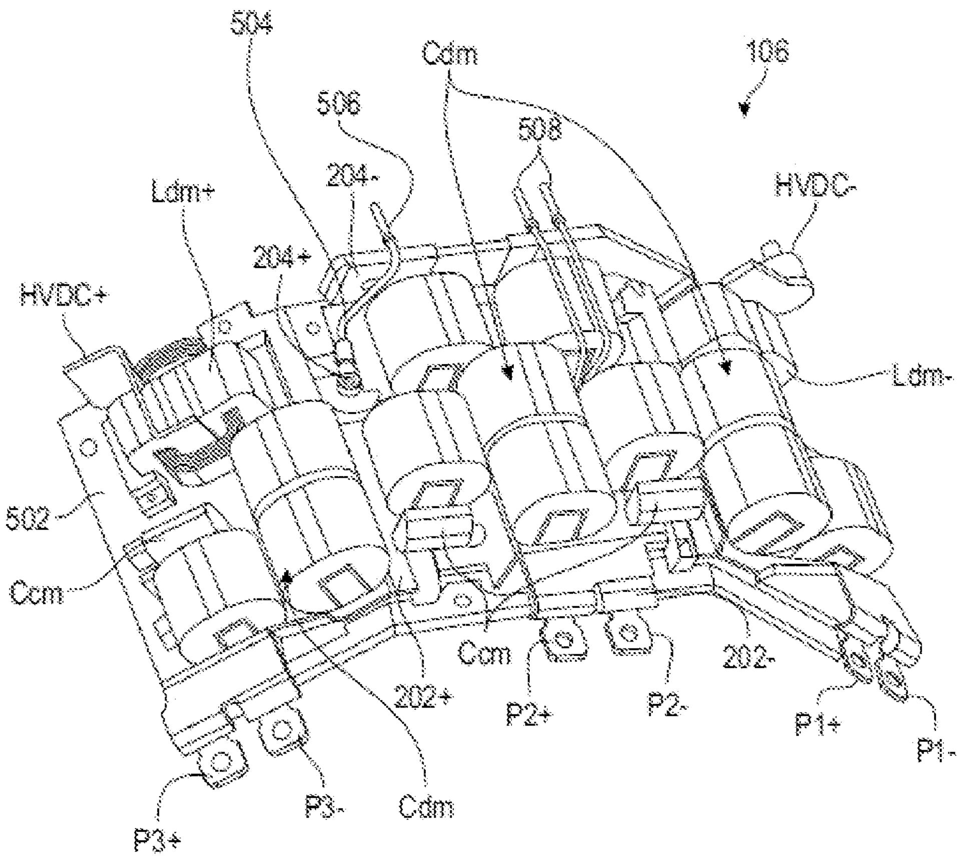
504
Cdm
106
506
508
204-
HVDC-
Ldm+
204+
HVDC+
Ldm-
502
Ccm
Ccm
202+
P2+
P2-
202-
P1+
P1-
P3-
Cdm
P3+

FIG. 5

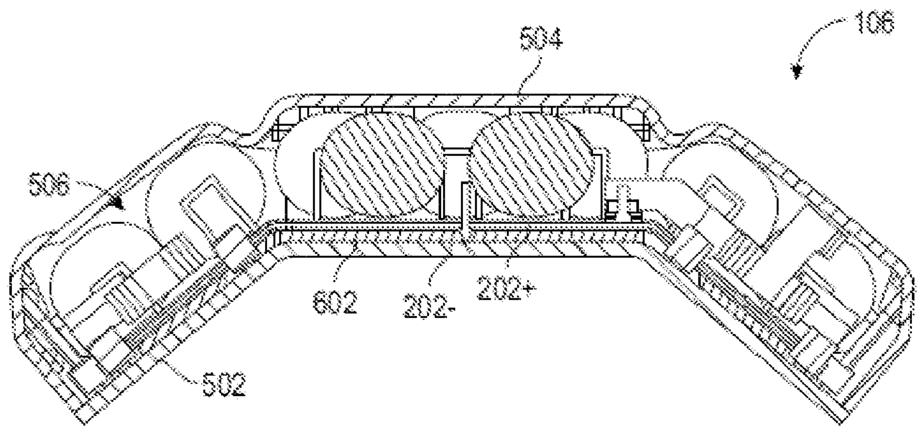

FIG. 6

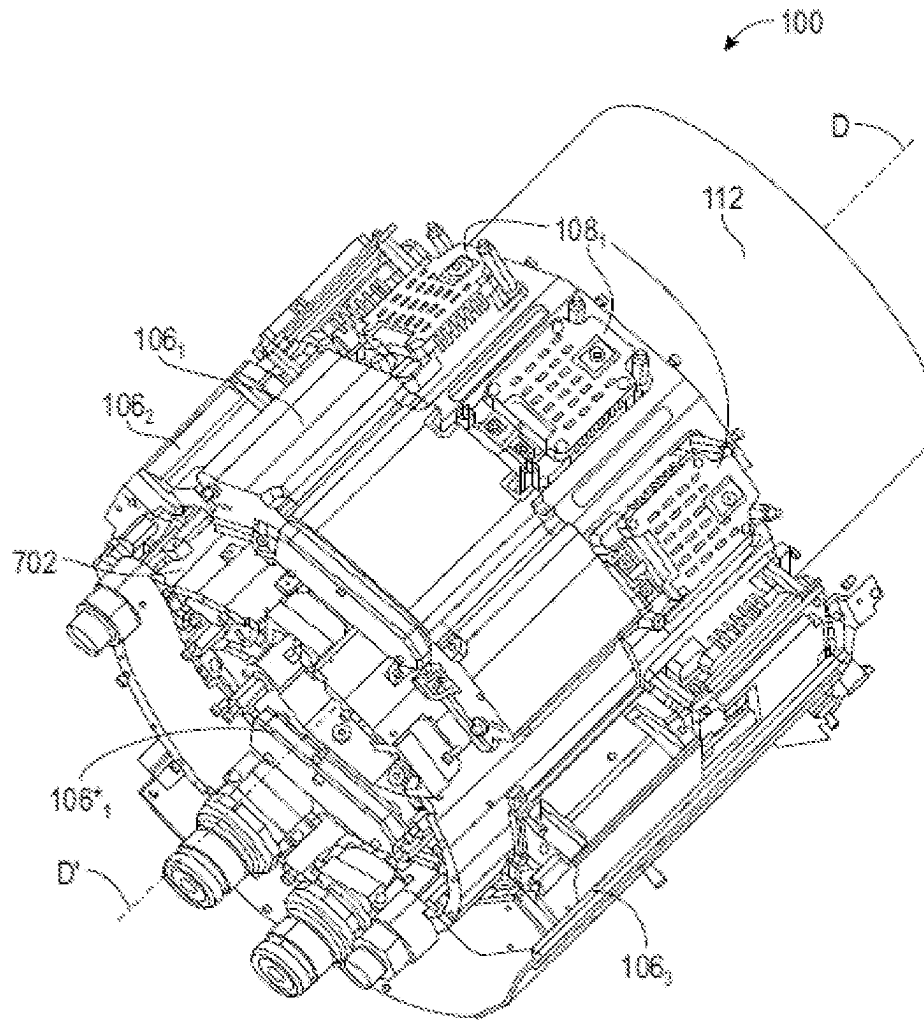

FIG. 7

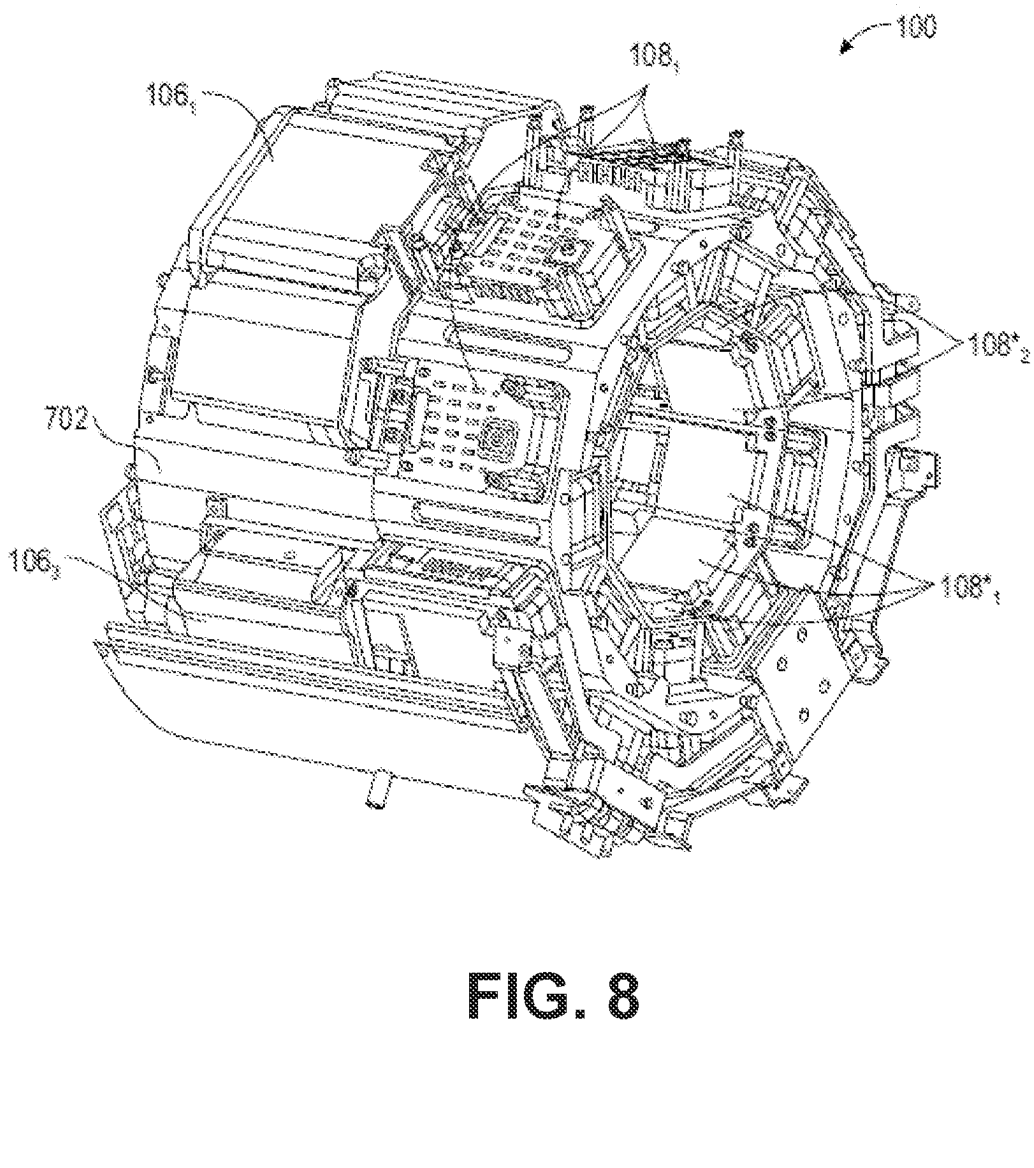

FIG. 8

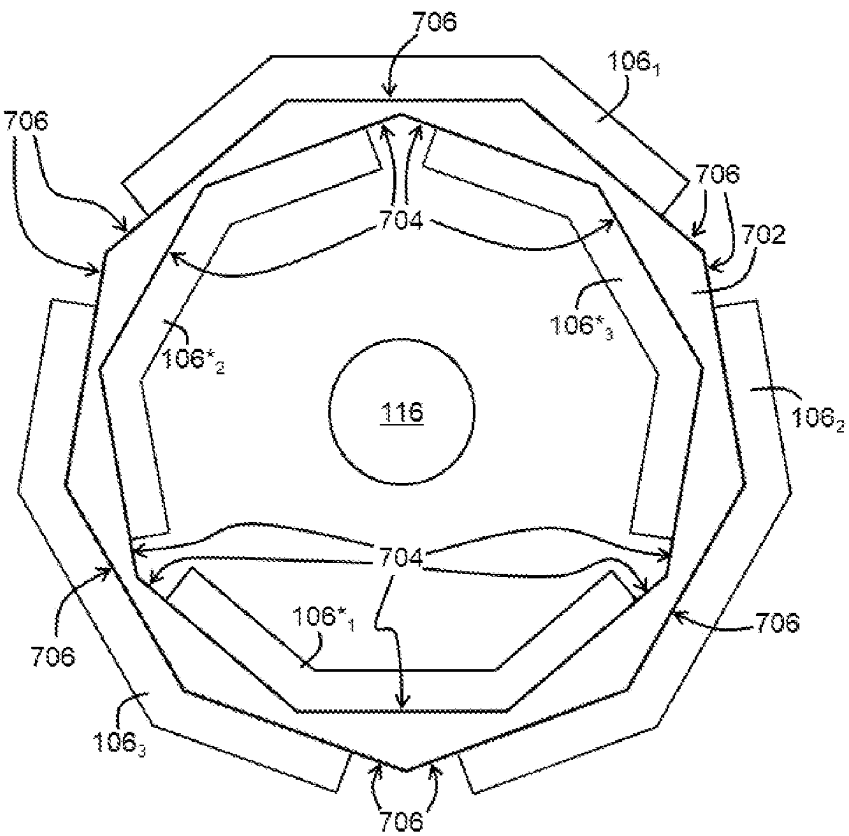
706
106_1
706
706
702
704
106^*_3
106^*_2
116
106_2
704
106^*_1
706
706
106_3
706

FIG. 9

DC VOLTAGE FILTER FOR AN INPUT OF AN INVERTER OF AN ELECTRIC MACHINE FOR PROPELLING AN AIRCRAFT

TECHNICAL FIELD OF THE INVENTION

The present invention relates to a DC voltage filter for an inverter input of an electric machine for propelling an aircraft. It also concerns a voltage converter comprising such a filter, an electric drive comprising such a voltage converter and an aircraft comprising such an electric drive.

Technological Background

The electric or hybrid propulsion aircraft, whether conventional, vertical take-off and landing (VTOL) or short take-off and landing (STOL), represent a market with significant prospects and demand for the intra-urban and inter-urban transport for transporting goods and people.

The electric drive is carried out by one or more electric motors, the number varying according to the architecture.

Generally speaking, the power is transmitted from a DC voltage source to the electric motor by a single inverter, and the latter comprises an input filter comprising, in particular, a high-value capacitor that is bulky.

It may be desirable to provide a device that can be compactly integrated into the electric machine.

The prior art relating to the electric motors and the capacitor of the input filter comprises the documents US 2020 343824 A1, US 2012 0274158 A1, EP 1 418 660 B1, U.S. Pat. No. 9,620,292 B2 and US 2010 0327680 A1.

SUMMARY OF THE INVENTION

A DC voltage filter is therefore proposed for an inverter input of an electric machine for propelling an aircraft, comprising:

- a positive busbar and a negative busbar, respectively having two portions each having substantially parallel folds so as to define at least three planes following one another and inclined with respect to one another, one of the portions, referred to as internal, being nested in the other, referred to as external, so that each plane of one extends substantially parallel opposite an associated plane of the other; and
- differential mode capacitors connected between the two portions.

Optionally, the positive busbar and the negative busbar comprise a positive input terminal and a negative input terminal respectively, and/or at least one positive output terminal and at least one negative output terminal respectively. Capacitive input filters are generally provided between global input terminals and the input terminals of the busbars. To avoid a confusion with the global input terminals, the input terminals of the busbars are referred to hereafter as "intermediate".

Optionally, more differential mode capacitors are connected between the central pair or pairs of planes than between the two peripheral pairs of planes.

Optionally, the differential mode capacitors are arranged on an external face of the external portion and/or on an internal face of the internal portion.

Also proposed is a voltage converter comprising:

- at least one power module; and
- a filter according to the invention, each power module being connected to a respective pair of positive and negative output terminals.

An electric drive is also proposed, comprising:

- a voltage converter according to the invention;
- an electric motor;
- an output shaft secured to a rotor of the electric motor, the input filter being arranged around the output shaft so that the portions of the busbars surround the output shaft.

Optionally, several voltage converters are distributed around the output shaft.

Also optionally, the electric drive further comprises a casing surrounding the output shaft and having a plurality of internal or external planar faces in succession around the output shaft, the input filter being pressed against the planar faces of the casing so that each plane of the lower portion extends substantially parallel opposite an associated planar face of the casing.

An aircraft comprising an electric drive according to the invention is also proposed.

BRIEF DESCRIPTION OF THE FIGURES

The invention will be better understood with the aid of the following description, given only by way of example and made with reference to the attached drawings in which:

FIG. **1** is a functional view of an example of an electric drive according to the invention for an aircraft,

FIG. **2** is an electrical diagram of an example of the input filter of a voltage converter of the electric drive of FIG. **1**,

FIG. **3** is a three-dimensional view of the busbars of the input filter in FIG. **2**, separated from each other,

FIG. **4** is a three-dimensional view of the busbars of the input filter in FIG. **2**, joined together,

FIG. **5** is a three-dimensional view of the input filter shown in FIG. **2**,

FIG. **6** is a cross-sectional view of the input filter in FIG. **2**,

FIG. **7** is a three-dimensional view of an example of a part of an electric drive according to the invention for an aircraft,

FIG. **8** is a three-dimensional view of the part of the electric drive of FIG. **7**, from a different angle, and

FIG. **9** is a cross-sectional view of the part of the electric drive in FIG. **7**.

DETAILED DESCRIPTION OF THE INVENTION

With reference to FIG. **1**, an example of an electric drive **100** for an aircraft will now be described.

The electric drive **100** firstly comprises a DC voltage source **102** designed to supply a DC voltage U. The DC voltage source **102** comprises, for example, one or more batteries.

The electric drive **100** also comprises at least one inverter $\mathbf{104}_{1\text{-}3}$ connected by a DC power harness **105** to the DC voltage source **102** and each designed to convert the DC voltage U into a respective AC voltage $V_{1\text{-}3}$. Each AC voltage $V_{1\text{-}3}$ is, for example, a polyphase voltage, in particular a three-phase voltage.

Each inverter $\mathbf{104}_{1\text{-}3}$ comprises in particular an input filter $\mathbf{106}_{1\text{-}3}$ and, for each input filter, one or more power modules, each designated by the global reference $\mathbf{108}_{1\text{-}3}$ and designed to respectively supply the different phase or phases of the AC voltage $V_{1\text{-}3}$ from the DC voltage U after filtering by the input filter $\mathbf{106}_{1\text{-}3}$. If the AC voltage $V_{1\text{-}3}$ is three-phase, for example, three power modules $\mathbf{108}_{1\text{-}3}$ are provided.

Each inverter **104** also comprises control electronics $\mathbf{110}_{1\text{-}3}$ designed to control the power modules $\mathbf{108}_{1\text{-}3}$.

Using several inverters **104** in parallel allows to reduce the electrical power passing through each of them.

The electric drive **100** also comprises an electric motor **112** with a stator and a rotor (not shown). The electric motor **112** is, for example, a permanent magnet synchronous motor.

The stator has stator windings (not shown) electrically connected to the power modules $\mathbf{108}_{1\text{-}3}$ by a polyphase power harness **113** to receive the phases of the AC voltages $V_{1\text{-}3}$ respectively. In response to the AC voltages $V_{1\text{-}3}$, the stator is designed to generate a rotating magnetic field.

The rotor, for example with a permanent magnet, is designed to be driven in rotation by the rotating magnetic field, in order to drive, for example, a propeller **114** of the electric drive **100**, to which the rotor is mechanically connected via an output shaft **116** of the electric drive **100**.

Preferably, as shown in FIG. **1**, a second electrical supply path is provided with elements similar to those of the first path. These similar elements carry the same reference numbers as those in the first path, with the addition of an asterisk "*". The two paths are preferably independent of each other so that a fault on one path does not cause a fault on the other. In this way, in the event of the loss of a complete path, a minimum supply via the other path can be guaranteed. In particular, this second path preferably comprises a second DC voltage source **102*** (independent of the first **102**) and inverters $\mathbf{104^*}_{1\text{-}3}$.

In the example shown, each path comprises three inverters in parallel, each capable of supplying up to 90 kW of electrical power. The electric motor **112** is designed to supply 500 kW of mechanical power. To achieve this, each path supplies 250 kW of electrical power (83.3 kW per inverter on that path).

With reference to FIG. **2**, one of the input filters $\mathbf{106}_{1\text{-}3}$, $\mathbf{106^*}_{1\text{-}3}$ will now be described in more detail. In FIGS. **2** to **6** it will be designated simply by the reference **106**, the inverter to which this input filter **106** belongs will be designated simply by the reference **104**, and the power modules connected to this filter will be designated simply by the reference **108**.

The input filter **106** firstly comprises a positive input terminal HVDC+ and a negative input terminal HVDC− designed to receive the DC voltage U.

The input filter **106** also comprises two differential mode inductors Ldm+, Ldm− respectively connected to the input terminals HVDC+, HVDC−.

The input filter **106** also comprises a positive busbar **202+** and a negative busbar **202−**, respectively having a positive intermediate terminal **204+** and a negative intermediate terminal **204−** respectively connected to the differential mode inductors Ldm+, Ldm−.

The differential mode inductors Ldm+, Ldm− have a value that ensures a network stability so that the resonance of the filter **106** is independent of the cable length (as far as possible). For example, each differential mode inductor Ldm+, Ldm− is between 2 and 3 pH, for example 2.5 pH.

The positive busbar **202+** also has at least one positive output terminal P1+, P2+, P3+, in particular one for each power module **108**.

Similarly, the negative busbar **202−** has at least one negative output terminal P1−, P2−, P3−, in particular one for each of the power modules **108** associated with this input filter **106**.

The busbars **202+**, **202−** are rigid electrical conductors, preferably laminated to reduce the switching loop inductor. Reducing the switching loop inductor is important when using power components in order to reduce the losses associated with switching the power modules **108** and to avoid over-voltages at the electric motor input **112**.

The input filter **106** also comprises several differential mode capacitors Cdm connected between the busbars **202+**, **202−**. The number of differential mode capacitors Cdm is preferably high, to allow them to be distributed, as will be explained later. Preferably there should be at least ten of them.

These differential mode capacitors Cdm together have a capacitance value chosen so as to reduce the variation in the DC voltage U and to ensure a sufficient network quality to support the currents of the inverter **104**. For example, the equivalent capacitor is between 100 and 200 ρF. For example, each capacitor has a value of 12.6 μF for a total of 151 μF for twelve differential mode capacitors Cdm in parallel.

The differential mode capacitors Cdm used are preferably polypropylene type capacitors whose service life depends on the voltage level and temperature. In power equipment, the service life is often given by the service life of the capacities used. So, to increase or guarantee a long service life, it is preferable to optimise the cooling of the differential mode capacitors Cdm.

The input filter **106** also comprises, for each output terminal, a common mode capacitor Cdm connected to the busbar to which that output terminal belongs.

The input filter **106** also comprises a device for measuring the current in at least one of the busbars **202+**, **202−**. In the example described, the current measuring device is designed to measure the current flowing in the negative busbar **202−** and comprises, for example, a resistor **206**.

With reference to FIGS. **3** and **4**, the busbars **202+**, **202−** will now be described in more detail.

The busbars **202+**, **202−** respectively comprise two portions **302+**, **302−** each having a curved shape, in a single orientation of curvature, and designed to nest into each other so as to extend at a substantially constant distance from each other.

In the example shown, the negative busbar **202−** comes to nest in the positive busbar **202+**. The positive busbar **202+** will hereinafter be referred to as external, while the negative busbar **202−** will hereinafter be referred to as internal. Of course, in other embodiments, the negative busbar **202−** could be external and the positive busbar **202+** could be internal.

The curvature ofthe portions **302+**, **302−** is obtained by the fact that each portion **302+**, **302−** has substantially parallel folds in order to define planes which follow one another and are inclined with respect to one another, the inclination always being in the same orientation (no zigzag). In the example shown, each portion **302+**, **302−** comprises three inclined planes **304+**, **306+**, **308+** and **304−**, **306−**, **308−**.

Once the busbars **202+**, **202−** are nested together, each plane of one of the busbars **202+**, **202−** extends substantially parallel opposite an associated plane of the other of the busbars **202+**, **202−**. More specifically, in the example shown, the planes **304+**, **304−** extend opposite each other, as do the planes **306+**, **306−** and the planes **308+**, **308−**.

The busbars **202+**, **202−** are preferably covered with an electrically insulating film **402**, which is shown transparently in FIG. **4**.

With reference to FIG. **5**, the differential mode capacitors Cdm are connected between the two portions **302+**, **302−** of the busbars **202+**, **202−**. Preferably, the differential mode capacitors Cdm are arranged on an external face of the external busbar and/or on an internal face of the internal busbar. In the example shown, all the differential mode capacitors are arranged on the external face of the positive busbar **202+**.

In particular, each pair of planes facing the busbars **202+**, **202−** carries at least two differential mode capacitors Cdm. Preferably, the central pair of planes (the planes **306+**, **306−** in the example shown) carries more differential mode capacitors Cdm than the peripheral pairs of planes (the planes **304+**, **304−** and **308+**, **308−** in the example described).

Thus, in the example shown, the pair of planes **306+**, **306−** carries six differential mode capacitors Cdm, while each of the pairs of planes **304+**, **304−** and **308+**, **308−** carries three differential mode capacitors Cdm.

In addition, the input filter **106** preferably comprises a base plate **502** carrying the bus plates **202+**, **202−**. In particular, the base plate **502** is itself curved to nest into the internal busbar and extend at a substantially constant distance from the latter. In the example shown, the base plate **502** comprises three planes extending parallel opposite to the planes **304+**, **306+**, **308+** of the negative busbar **202−**.

The differential mode inductors Ldm+, Ldm− are, for example, carried by the base plate **502**. Preferably, a thermal interface is provided between each differential mode inductor Ldm+, Ldm− and the base plate **502** to optimise the cooling towards the base plate **502**.

The input filter **106** may also comprise, carried by the busbars **202+**, **202−** and/or by the base plate **502**, a device **504** for measuring the voltage U and a connector **506** for transmitting the voltage measurement.

The input filter **106** may further comprise a device (not visible) for measuring a temperature of the input filter **106**, in particular a temperature of one of the differential mode capacitors Cdm and connectors **508** for transmitting the temperature measurement.

For example, the resistor **206** used to measure current comprises a current shunt **510** with two pins **512** between which a voltage is measured to deduce the current.

All or some of these measurements are sent to the control electronics **110** of the power modules **108**, to be used to control the switches of the power modules **108**. To do this, the control electronics **110** comprises, for example, a module for processing and digitising the measurements, this processing module receiving the connectors **506**, **508** and the pins **512**.

With reference to FIG. **6**, an electrically insulating layer **602** is provided between the internal busbar (the negative busbar **202−** in the example described) and the base plate **502**.

The electrical insulating layer **602** should preferably be able to withstand 1000 V under all altitude and temperature conditions encountered by the aircraft, between the two busbars **202+**, **202−** and the base plate **502** forming an electrical mass.

The input filter **106** also comprises a housing **504** which comes into contact with the base plate **502** and covers the differential mode capacitors Cdm, as well as, for example, the differential mode inductors Ldm and/or the common mode capacitors Ccm. The housing **504** is for example filled with a thermal insulating material **506**, such as a resin, so that the heat generated by the differential mode capacitors Cdm and the differential mode inductors Ccm is dissipated towards the busbars **202+**, **202−** and then through the base plate **502**. Thus, thanks to the thermal insulating material **506**, a direct heat transfer between the differential mode inductors Ldm and the differential mode capacitors Cdm is limited. This is because the differential mode inductors Ldm generally heat up much more than the differential mode capacitors Cdm, and the latter generally cannot withstand very high temperatures. In this way, it is possible to place these components close to each other, and thus achieve a good compactness, without the components that heat a lot raising the temperature of the components that heat less and cannot withstand too high a temperature. For example, the thermal insulating material **506** has a thermal conductivity of less than 0.5 W/m/K.

In addition, in the event of a problem with a component, the thermal insulating material **506** is designed to contain an explosion of this component.

With reference to FIGS. **7** to **9**, an example of integration of the electric drive **100** will now be described.

In this example, each power transmission path comprises three inverters $\mathbf{104}_{1\text{-}3}$, $\mathbf{104*}_{1\text{-}3}$. The electric drive **100** therefore comprises six input filters $\mathbf{106}_{1\text{-}3}$, $\mathbf{106*}_{1\text{-}3}$.

The output shaft **116** (not shown in FIGS. **7** and **8**) secured to the rotor of the electric motor **112** extends along an axis of rotation DD'.

The electric drive **100** comprises a casing **702** surrounding the output shaft **116** and having a number of planar faces, internal **704** and external **706** in succession around the output shaft **116**. Preferably, the internal **704** and external **706** faces have the same number and are substantially angularly offset around the axis of rotation DD' by half a planar face. In the example shown, the casing **702** has nine internal planar faces **704** and nine external planar faces.

The input filters $\mathbf{106}_{1\text{-}3}$, $\mathbf{106}_{1\text{-}3}$* are distributed around the axis of rotation DD' with their curvature around the axis of rotation DD'. More specifically, in the example shown, the input filters $\mathbf{106}_{1\text{-}3}$ of the first path are pressed against the external faces **706**, while the input filters $\mathbf{106*}_{1\text{-}3}$ of the second path are pressed against the internal faces **704**. In this way, each input filter $\mathbf{106}_{1\text{-}3}$, $\mathbf{106}^{*1\text{-}3}$ is pressed against planar faces **704** or **706** of the casing **702** so that each base plate **502** (and therefore each plane of the portion of the lower busbar) extends substantially parallel opposite an associated planar face of the casing **702**.

In a similar way, the power modules $\mathbf{108}_{1\text{-}3}$, $\mathbf{108}_{1\text{-}3}$* are for example distributed around the axis of rotation DD' and placed between the input filters $\mathbf{106}_{1\text{-}3}$, $\mathbf{106}_{1\text{-}3}$* and the electric motor **112**. More precisely, each power module $\mathbf{108}_{1\text{-}3}$, $\mathbf{108*}_{1\text{-}3}$ is pressed against an internal **704** or external **706** planar face of the casing **702**, in particular against one of the planar faces on which the input filter $\mathbf{106}_{1\text{-}3}$, $\mathbf{106}_{1\text{-}3}$* of this power module $\mathbf{108}_{1\text{-}3}$, $\mathbf{108}_{1\text{-}3}$* is pressed. The angular offset between the internal faces **704** and the external faces **706** allows the power modules $\mathbf{108}_{1\text{-}3}$, $\mathbf{108}_{1\text{-}3}$* to be placed in series one after the other along the same cooling circuit, the latter alternately cooling a power module $\mathbf{108}_{1\text{-}3}$* on an internal face **704**, then that on an external face **706**, and so on.

The inlet filters $\mathbf{106}_{1\text{-}3}$, $\mathbf{106}_{1\text{-}3}$* are compactly integrated into a small, rounded volume. In addition, the length of the polyphase harnesses $\mathbf{113}_{1\text{-}3}$, $\mathbf{113}_{1\text{-}3}$* is very short, which allows to reduce the parasitic inductors between the differential mode capacitors Cdm and the power modules $\mathbf{108}_{1\text{-}3}$, $\mathbf{108*}_{1\text{-}3}$.

In this way, the heat dissipation towards the casing **702** is facilitated by the fact that the busbars **202+**, **202−** are pressed against the casing **702** (through the base plate **502** in the example described) over a large surface area, due to their curved shape which complements that of the casing **702**.

It should be noted that the invention is not limited to the embodiments described above. In fact, it will appear to the person skilled in the art that various modifications can be made to the above-described embodiments, in the light of the teaching just disclosed.

In particular, the input filter could be purely capacitive and not comprise differential mode inductors.

In the foregoing detailed presentation of the invention, the terms used should not be interpreted as limiting the invention to the embodiments exposed in the present description, but should be interpreted to include all equivalents the anticipation of which is within the reach of the person skilled in the art by applying his general knowledge to the implementation of the teaching just disclosed.

The invention claimed is:

1. A filter of DC voltage (U) for an inverter input of an electric machine for propelling an aircraft, comprising:
- a positive busbar and a negative busbar; and
- differential mode capacitors (Cdm) connected between two respective portions of the positive busbar and the negative busbar;
- wherein each of the two portions each have parallel folds so as to define at least three planes following one another and inclined with respect to one another, one of the two portions being an internal portion and the other of the two portions being an external portion, the internal portion being nested in the external portion so that each plane of one of the two portions extends parallel opposite an associated plane of the other of the two portions.

2. The filter of claim **1**, wherein more differential mode capacitors (Cdm) are connected between central pair or pairs of planes than between two peripheral pairs of planes.

3. The filter according to claim **1**, wherein the differential mode capacitors (Cdm) are arranged on an external face of the external portion and/or on an internal face of the internal portion.

4. A voltage converter comprising:
- at least one power module performing an inverter function; and
- the filter according to claim **1**, each power module being connected to a respective pair of positive and negative output terminals.

5. An electric drive comprising:
- the voltage converter according to claim **4**;
- an electric motor;
- an output shaft secured to a rotor of the electric motor, the filter being an input filter arranged around the output shaft so that the portions of the busbars surround the output shaft.

6. An aircraft comprising the electric drive according to claim **5**.

7. The electric drive according to claim **5**, further comprising a plurality of voltage converters distributed around the output shaft.

8. The electric drive according to claim **7**, further comprising a casing surrounding the output shaft and having a plurality of internal or external planar faces in succession around the output shaft, the input filter being pressed against the planar faces of the casing so that each plane of a lower portion of the casing extends parallel opposite an associated planar face of the casing.

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